

Reliable Accuracy Estimates from k -Fold Cross Validation

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Abstract—It is popular to evaluate the performance of classification algorithms by k -fold cross validation. A reliable accuracy estimate will have a relatively small variance, and several studies therefore suggested to repeatedly perform k -fold cross validation. Most of them did not consider the correlation among the replications of k -fold cross validation, and hence the variance could be underestimated. The purpose of this study is to explore whether k -fold cross validation should be repeatedly performed for obtaining reliable accuracy estimates. The dependency relationships between the predictions of the same instance in two replications of k -fold cross validation are first analyzed for k -nearest neighbors with $k = 1$. Then, statistical methods are proposed to test the strength of the dependency level between the accuracy estimates resulting from two replications of k -fold cross validation. The experimental results on 20 data sets show that the accuracy estimates obtained from various replications of k -fold cross validation are generally highly correlated, and the correlation will be higher as the number of folds increases. The k -fold cross validation with a large number of folds and a small number of replications should be adopted for performance evaluation of classification algorithms.

Index Terms—Dependency relationship, k -fold cross validation, reliable estimate, replication, variance

1 INTRODUCTION

THE performance of classification algorithms is generally evaluated by k -fold cross validation. A data set is first randomly divided into k disjoint folds that have approximately the same number of instances. Then every fold in turn plays the role for testing the model induced from the other $k-1$ folds. Since the partition is random, the variance of the accuracy estimates can be large for statistical inference [1]. Bengio and Grandvalet [2] showed that the variance can be represented as a linear combination of three components. Several studies suggested to perform k -fold cross validation repeatedly to obtain reliable accuracy estimates for statistical comparison. For example, 10-fold cross validation is recommended to perform ten times in [3], [4]. Vanwinckelen and Blockeel [5] and Kim [6] also pointed out that repeating k -fold cross validation can stabilize the variability of accuracy estimates. The nonparametric methods proposed by Demšar [7] for the performance evaluation of classification algorithms rely on the reliable estimates obtained from repeating stratified random splits. The six books summarized in [8] do not have consistent conclusion about whether k -fold cross validation should be repeatedly performed.

The training sets in two iterations of k -fold cross validation are overlapping when the number of folds is larger than two. The accuracy estimates resulting from k -fold cross validation can thus be dependent to interfere the estimation

of variance. To avoid the overlapping of training sets, Dietterich [9] proposed a hypothesis test for performance evaluation based on repeating two-fold cross validation five times, denoted by 5×2 cross validation. Alpaydm [10] extended this approach to design an F test. A similar approach for estimating the accuracies of classification algorithm is the blocked 3×2 cross validation proposed by Wang et al. [11]. Wang et al. [12] studied the quality of data partitioning in $m \times 2$ cross validation.

There are thus two main reasons in repeating k -fold cross validation: having more observations to obtain a reliable accuracy estimate and reducing the dependency among accuracy estimates. When k -fold cross validation is applied on a data set repeatedly, the fold accuracies resulting from various replications can be dependent. This dependency will have impact on calculating the variance of accuracy estimates, and hence it is an essential issue in evaluating the performance of classification algorithms.

Kohavi [13] suggested that the number of folds should not be less than ten for variance estimation, and that repeated runs can reduce variance when the number of folds is between two and five. Rodriguez et al. [14] recommended that the number of folds should be five or ten to have a smaller bias. The lowest variance can be achieved when the number of folds is two. The conclusions of those studies are not completely consistent, and none of them explicitly considered the dependency among various replications of k -fold cross validation. In addition, the way to calculate the variance for performance evaluation in previous studies are not all the same. Hence, when k -fold cross validation is repeatedly performed, the problems that will be investigated in this study are:

- Analyze possible ways of calculating variance estimates.

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- Explore the dependency relationships among the accuracy estimates obtained from various replications.
- Make recommendation about the number of folds and the number of replications in applying k -fold cross validation to obtain reliable accuracy estimates.

This study will first analyze the prediction results of k -nearest neighbors with $k = 1$ to show that the accuracy estimates obtained from various replications of k -fold cross validation are dependent. Then statistical methods are proposed to test the strength of the dependency level among the accuracy estimates resulting from various replications. The recommendation for applying k -fold cross validation to have reliable accuracy estimates will be given based on the experimental results on twenty data sets.

The remainder of this paper is organized as follows. Section 2 presents three ways to compute the variance of the accuracy estimates resulting from various replications of k -fold cross validation. Section 3 introduces k -nearest neighbors with $k = 1$ to investigate the dependency relationships among the predictions in various replications. The statistical methods for testing the dependency level among the accuracies obtained from various replications are proposed in Section 4, and the experimental results of twenty data sets are shown in Section 5 to demonstrate that the accuracy estimates resulting from various replications are highly correlated. The experimental results are discussed in Section 6 to make recommendations about the application of k -fold cross validation. Section 7 presents the conclusions and future directions of this research.

2 ALTERNATIVES FOR COMPUTING VARIANCE

A reliable accuracy estimate has a relatively small variance. Several studies introduced in Section 1 suggested that k -fold cross validation can be repeated to calculate the variance of accuracy estimates, while their ways for computing the variance are not all the same. This section presents three possible ways for computing the variance from the classification results of k -fold cross validation to identify whether their variance estimates can be used for statistical inference.

In a replication of k -fold cross validation, a data set D that has n instances collected by the simple random sampling is first randomly divided into k disjoint folds such that every fold has m instances, and hence $n = k \times m$. Then every fold is in turn used to test the model induced from the other $k - 1$ folds by a classification algorithm L . Let x_{iju} be the prediction of the u^{th} instance in the j^{th} fold of the i^{th} replication, and $x_{iju} = 1$ if the prediction is correct, and $x_{iju} = 0$ otherwise. Then the accuracy obtained from the j^{th} fold of the i^{th} replication is $\bar{p}_{ij} = \sum_{u=1}^m x_{iju} / m$. The large-sample conditions for $\bar{p}_{ij}$ to follow a normal distribution with mean p and variance $p(1-p)/m$ are: the numbers of correct and wrong predictions in this fold are both larger than or equal to five [8], where p is the actual accuracy of algorithm L on the population corresponding to data set D . A set of fold accuracies $\{\bar{p}_{ij}, i = 1, 2, \dots, l, j = 1, 2, \dots, k\}$ obtained from l replications will be used to calculate the variance of sample accuracy. Define $\bar{x}_i = \sum_{j=1}^k \bar{p}_{ij} / k$ and $\bar{\bar{x}} = \sum_{i=1}^l \sum_{j=1}^k \bar{p}_{ij} / lk$ to be the sample mean accuracy for replication i and overall sample mean accuracy, respectively. The fold accuracies obtained in the same replication are assumed to be independent [15].

Theorem 1. Let s_{ind}^2 be the sample variance calculated as $s_{ind}^2 = \sum_{i=1}^l \sum_{j=1}^k (\bar{p}_{ij} - \bar{\bar{x}})^2 / (lk - 1)$ when k -fold cross validation is repeated l times. If the fold accuracies obtained from various replications are independent, then $E(s_{ind}^2) = p(1-p)/m$.

Proof. Since $E(\bar{p}_{ij}) = p$ for $i = 1, 2, \dots, l$ and $j = 1, 2, \dots, k$, we have

$$\begin{aligned} E(s_{ind}^2) &= \frac{1}{lk-1} \sum_{i=1}^l \sum_{j=1}^k E(\bar{p}_{ij}^2 - 2\bar{p}_{ij}\bar{\bar{x}} + \bar{\bar{x}}^2) \\ &= \frac{1}{lk-1} \sum_{i=1}^l \sum_{j=1}^k [E(\bar{p}_{ij}^2) - 2E(\bar{p}_{ij}\bar{\bar{x}}) + E(\bar{\bar{x}}^2)] \\ &= \frac{1}{lk-1} \sum_{i=1}^l \sum_{j=1}^k [Var(\bar{p}_{ij}) + E(\bar{p}_{ij})^2 - 2E(\bar{p}_{ij}\bar{\bar{x}}) + Var(\bar{\bar{x}}) + E(\bar{\bar{x}})^2] \\ &= \frac{1}{lk-1} \sum_{i=1}^l \sum_{j=1}^k \left[Var(\bar{p}_{ij}) + p^2 - \frac{2}{lk} E\left(\bar{p}_{ij} \sum_{u=1}^l \sum_{v=1}^k \bar{p}_{uv}\right) + Var(\bar{\bar{x}}) + p^2 \right]. \end{aligned}$$

If the $\bar{p}_{ij}$ are all independent, then $Var(\bar{\bar{x}}) = Var(\bar{p}_{ij})/lk$ and $E(\bar{p}_{ij}\bar{p}_{uv}) = E(\bar{p}_{ij})E(\bar{p}_{uv}) = p^2$ for any $i \neq u$ or $j \neq v$. We therefore have

$$\begin{aligned} E(s_{ind}^2) &= \frac{1}{lk-1} \sum_{i=1}^l \sum_{j=1}^k \left\{ Var(\bar{p}_{ij}) + p^2 \right. \\ &\quad \left. - \frac{2}{lk} [(lk-1)p^2 + Var(\bar{p}_{ij}) + p^2] + \frac{1}{lk} Var(\bar{p}_{ij}) + p^2 \right\} \\ &= \frac{1}{lk-1} \sum_{i=1}^l \sum_{j=1}^k \left[Var(\bar{p}_{ij}) - \frac{2}{lk} Var(\bar{p}_{ij}) + \frac{1}{lk} Var(\bar{p}_{ij}) \right] \\ &= Var(\bar{p}_{ij}) = \frac{p(1-p)}{m}. \end{aligned}$$

□

Theorem 1 shows that when the fold accuracies obtained from various replications are independent, sample variance s_{ind}^2 is an unbiased estimator of fold variance $p(1-p)/m$. In this case, the variance of $\bar{\bar{x}}$ is expressed as

$$Var(\bar{\bar{x}}) = Var\left(\frac{\sum_{i=1}^l \sum_{j=1}^k \bar{p}_{ij}}{lk}\right) = \frac{\sum_{i=1}^l \sum_{j=1}^k Var(\bar{p}_{ij})}{l^2 k^2} = \frac{p(1-p)/m}{lk}.$$

When the estimate of $p(1-p)/m$ is s_{ind}^2 , the estimate of $Var(\bar{\bar{x}})$ is s_{ind}^2/lk . This indicates that the estimate of $Var(\bar{\bar{x}})$ decreases as the number of replications increases, and it will be zero as the value of l goes to infinite. This is the reason why several previous studies addressed that k -fold cross validation can be repeated to obtain reliable accuracy estimates. Since the size of data set D remains unchanged, it is not reasonable that the estimate of $Var(\bar{\bar{x}})$ can be reduced to zero by repeating k -fold cross validation. Vanwinckelen and Blockeel [5] also pointed out that the confidence intervals or significance resulting from repeated cross-validation can be misleading.

The second way averages the sample variance calculated from each replication, and the formula can be expressed as $\bar{s}^2 = \sum_{i=1}^l [\sum_{j=1}^k (p_{ij} - \bar{x}_i)^2 / (k-1)] / l$. It is obvious that $\bar{s}^2$ is an unbiased point estimator of $p(1-p)/m$. Since sample variances are calculated replication by replication, this approach need not assume that the fold accuracies resulting from various replications are independent. The value of $\bar{s}^2$ will converge to $p(1-p)/m$ as the value of l increases. The

TABLE 1
The Experimental Results for Example 1

Replication	Fold					Mean $\bar{x}$	Variance $s_{\bar{x}}^2$
	1	2	3	4	5		
1	0.78	0.77	0.72	0.80	0.78	0.7700	0.000900
2	0.78	0.80	0.80	0.72	0.80	0.7800	0.001200
3	0.82	0.80	0.74	0.80	0.74	0.7800	0.001400
4	0.79	0.74	0.76	0.78	0.76	0.7660	0.000380

variance for statistical inference is set to be $\bar{s}^2/k$ in this case. Since the sample variance obtained from single replication of k -fold cross validation can be large, several previous studies employs this approach to obtain a relatively stable variance estimate. Computational efficiency is the major concern of this alternative.

The probability for an instance to be correctly classified by algorithm L equals p . If the n instances in D are classified independently, then the resulting accuracy can be assumed to follow a normal distribution with mean p and variance $p(1-p)/n$ when the numbers of correct and wrong predictions are both larger than or equal to five. We can perform k -fold cross validation only once, and the resulting accuracy $\bar{x}_1$ is an unbiased point estimator of p . The variance for statistical inference is therefore set to be $\bar{x}_1(1-\bar{x}_1)/n$. Since $s_{\bar{x}}^2 = \sum_{j=1}^k (\bar{p}_{1j} - \bar{x}_1)^2 / (k-1)$ is an unbiased point estimator of $p(1-p)/m$, $p(1-p)/n$ can also be estimated by $s_{\bar{x}}^2/k$. Estimating $p(1-p)/n$ by $s_{\bar{x}}^2/k$ can reflect the variability among fold accuracies, while it is more unstable with respect to estimate $\bar{x}_1(1-\bar{x}_1)/n$. The dependency relationships among various replication accuracies is not an issue in this case, because k -fold cross validation is performed only once. Since two-fold cross validation is tended to have a high bias, it is generally repeated to obtained a reliable accuracy estimate $\bar{x}$ for calculating variance by expression $\bar{x}(1-\bar{x})/n$ [5], [13], [14]. As pointed by Kohavi [13], when a classification algorithm is relatively stable, variance estimate $\bar{x}(1-\bar{x})/n$ does not depend on the number of folds. Wong and Yang [15] also addressed that $\bar{x}(1-\bar{x})/n$ is not sensitive to the value of $\bar{x}$.

Example 1. Suppose that 5-fold cross validation is employed to evaluate the performance of a classification algorithm L on a data set D that contains 1,000 instances. The experimental results are shown in Table 1 when the number of replications is four. The overall sample mean is $\bar{x} = \sum_{i=1}^4 \sum_{j=1}^5 \bar{p}_{ij} / 20 = 0.7740$. If the 20 fold accuracies are considered to be independent, then sample variance equals $s_{ind}^2 = \sum_{i=1}^4 \sum_{j=1}^5 (\bar{p}_{ij} - \bar{x})^2 / 19 = 0.000857$, and hence the estimate of $Var(\bar{x})$ is $0.000857/20 = 0.000043$. We can also average the sample variances of the four replications to obtain

$$\bar{s}^2 = (.000900 + .001200 + .001400 + .000380) / 4 = .000970,$$

and the estimate of $Var(\bar{x})$ is $0.000970/5 = 0.000194$. If 5-fold cross validation is performed only once, say replication 1, then $\bar{x}_1(1-\bar{x}_1)/n = 0.77 \times (1-0.77)/1,000 = 0.000177$. Since $s_{\bar{x}}^2 = 0.000900$, the variance estimate for statistical inference can also be calculated as $0.000900/5 = 0.000180$.

The values of s_{ind}^2 , $\bar{s}^2$, and $s_{\bar{x}}^2$ in Example 1 are close, while s_{ind}^2/lk is much smaller than the estimates of $Var(\bar{x})$ obtained from the other two approaches. This will be getting worse as the number of replications increases. When 5-fold cross validation is performed only once, the variance estimates calculated by $\bar{x}(1-\bar{x})/n$ corresponding to the four replications are 0.000177, 0.000172, 0.000172, and 0.000179, respectively. Variance estimator $\bar{x}(1-\bar{x})/n$ is thus not sensitive to the value of $\bar{x}$. The estimates $s_{\bar{x}}^2/5$ corresponds to the four replications are 0.000240, 0.000180, 0.000280, and 0.000076, respectively. This demonstrates that when k -fold cross validation is performed only once, estimating variance by $s_{\bar{x}}^2/k$ is relatively unstable.

3 K-NEAREST NEIGHBORS WITH $k = 1$

The main issue in estimating $Var(\bar{x})$ by s_{ind}^2/lk is that the correlations among the $\bar{p}_{ij}$ for $i = 1, 2, \dots, l$ and $j = 1, 2, \dots, k$ do not take into account in calculating s_{ind}^2 . As pointed out by Wong and Yang [15], the $\bar{p}_{ij}$ resulting from the same replication can be assumed to be independent. This implies that the correlations among the $\bar{p}_{ij}$ in various replications should not be ignored in estimating $Var(\bar{x})$. For any given testing instance, algorithm k -nearest neighbors finds its k closest instances in training data and takes their most common class value to be the prediction. Since k -nearest neighbors with $k = 1$, denoted as 1NN, has a relatively simple classification mechanism, it will be used to investigate the dependency among the accuracy estimates obtained from various replications of k -fold cross validation in this section. The results can help us to explore the dependency among the replications in the other algorithms.

Let a data set D collected by the simple random sampling have instances $e_1, e_2, \dots, e_n$, and let p be the actual accuracy of 1NN on the population corresponding to D . The j^{th} nearest neighbor of e_i in D is denoted as $e_{(j)}^i$. Suppose that k -fold cross validation is performed twice on this data set, and every fold has m instances; i.e., $n = k \times m$.

Theorem 2. Let a data set have $n = k \times m$ instances. When classification algorithm is 1NN, the probability for the predictions of an instance to be dependent in two replications of k -fold cross validation is $\sum_{j=1}^m (C_{m-j-1}^{n-j-1} / C_{m-1}^{n-1})^2$.

Proof. Since instances are collected independently, the two predictions of e_i will be dependent only when it is classified by the same instances in the two replications. In any replication, instance e_i will be classified by $e_{(j)}^i$ if e_i and $e_{(u)}^i$ for $u = 1, 2, \dots, j-1$ are in the same fold, and e_i and $e_{(j)}^i$ are in different folds. The probability for e_i to be classified by $e_{(j)}^i$ thus equals

$$P(e_i \text{ is classified by the } j^{th} \text{ nearest neighbor}) \\ = C_{m-j}^{(n-2)-(j-1)} / C_{m-1}^{n-1} = C_{m-j}^{n-j-1} / C_{m-1}^{n-1}.$$

Dividing a data set into k disjoint folds in different replications are independent, and every fold has m instances. The probability for e_i to be classified by the same instance in two replications is thus equal to $\sum_{j=1}^m (C_{m-j}^{n-j-1} / C_{m-1}^{n-1})^2$. $\square$

When the number of folds increases, the number of instances in a fold will decrease. An instance thus has a larger probability to be classified by the same instance in two

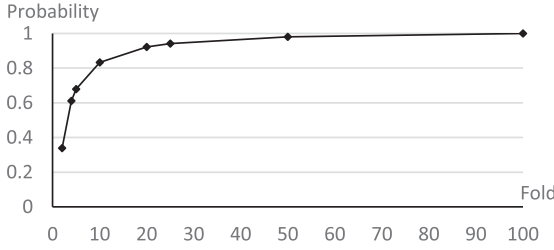


Fig. 1. The dependency probability for various number of folds in a data set with 100 instances.

replications. Consider instances e_i and $e_{(1)}^i$. The probability for these two instances to be located in different folds is $(n - m)/(n - 1)$. Since $n = k \times m$, the value of m is getting small as the value of k increases. In leave-one-out cross validation, $k = n$ and $m = 1$, and e_i is always classified by $e_{(1)}^i$. The predictions of an instance in two replications of leave-one-out cross validation must therefore be fully dependent.

Example 2. Suppose that a data set D has 100 instances. When the number of folds is five, every fold has 20 instances. By Theorem 2, the probability for the predictions of an instance to be dependent in two replications of 5-fold cross validation is $\sum_{j=1}^{20} (C_{20-j}^{100-j-1} / C_{20-1}^{100-1})^2 = 0.6784$. Fig. 1 shows this probability for $k = 2, 4, 5, 10, 20, 25, 50$, and 100. We can see from Fig. 1 that this probability increases as the number of folds becomes large.

Theorem 3. Suppose that k -fold cross validation is performed twice. Let e_i and e_j be two distinct instances in a data set having $n = k \times m$ instances, and let $e_{(u)}^i = e_i$ and $e_{(v)}^j = e_j$ for some $u \leq m$ and $v \leq m$. When classification algorithm is 1NN, let A be the event that e_i is classified by e_j in replication 1, and let B be the event that e_j is classified by e_i in replication 2. Then

$$P(A \cap B) = \frac{C_{m-v-1}^{m-u-1}}{C_{m-1}^{m-1}} \times \frac{C_{m-u-1}^{m-v-1}}{C_{m-1}^{m-1}}.$$

Proof. As argued in the proof of Theorem 2, $P(A) = C_{m-v-1}^{m-u-1} / C_{m-1}^{m-1}$ and $P(B) = C_{m-u-1}^{m-v-1} / C_{m-1}^{m-1}$. Since dividing instances into folds in various replications are independent, we have $P(A \cap B) = P(A) \times P(B)$. $\square$

When e_i is classified by e_j in replication 1, and e_j is classified by e_i in replication 2, these two predictions will be dependent. Since $n = k \times m$, the number of instances in a fold is getting small as the number of folds increases. By Theorem 3, the probability for the predictions of two distinct instances to be dependent in two replications thus increases as the number of folds increases. Theorem 2 also indicates that a large number of folds will increase the probability for the predictions of the same instance to be dependent in two replications. We therefore conclude that the number of folds will have impact on the dependency among the fold accuracies resulting from various replications in 1NN. However, the other algorithms do not have such a simple classification mechanism as 1NN to draw an analytical conclusion.

4 DEPENDENCY TESTS

As argued in the previous section, determining the dependency level among the accuracy estimates in various replications of k -fold cross validation analytically is complicated.

TABLE 2
The 2×2 Contingency Table for the Predictions in Two Replications of k -Fold Cross Validation

		Replication 2	
		Correct	Wrong
Replication 1	Correct	b_{11}	b_{10}
	Wrong	b_{01}	b_{00}

This section will introduce two statistical tests to study this dependency relationship. One is the independent test for binary random variables, and another is to test the dependency level of the predictions resulting from two replications of k -fold cross validation.

Since the partitions of instances into folds for various replications are independent, fold accuracies cannot be used to evaluate the dependency relationship among various replications. The dependency relationship is thus analyzed by the predictions of instances. Let x_{ij} be the prediction of the j^{th} instance in replication i for $i = 1, 2$ and $j = 1, 2, \dots, n$, and $x_{ij} = 1$ if the prediction is correct, and $x_{ij} = 0$ otherwise. Every x_{ij} is a binary random variable, and hence the predictions in two replications can be summarized in a 2×2 contingency table as shown in Table 2 in which b_{uv} represents the frequency of $x_{1j} = u$ and $x_{2j} = v$. Then the correlation coefficient ϕ between the predictions in two replications is estimated as

$$\bar{\phi} = \frac{b_{11}b_{00} - b_{10}b_{01}}{\sqrt{(b_{11} + b_{10}) \times (b_{01} + b_{00}) \times (b_{11} + b_{01}) \times (b_{10} + b_{00})}}.$$

When those predictions are independent, $n\bar{\phi}$ will follow a chi-squared distribution with one degree of freedom [16]. The statistic for testing hypothesis H_0 : the predictions of the same instance in two replications are independent is calculated as $\chi^2 = n\bar{\phi}$. When the significance level is α , the null hypothesis will be rejected if $\chi^2 \geq \chi_{\alpha,1}^2$.

The chi-squared test is to identify whether the predictions of the same instance in two replications can be assumed to be independent. It cannot be used to evaluate the dependency level of the accuracy estimates resulting from two replications, and the statistical test proposed in the following is designed for this purpose. Let the actual accuracy of a classification algorithm on the population corresponding to data set D be p . The predictions of an instance e_j in two replications can be categorized as $x_{1j} = x_{2j}$ or $x_{1j} \neq x_{2j}$.

Definition 1. For any classification algorithm evaluated by k -fold cross validation on a data set, let x_{ij} be the prediction of the j^{th} instance in replication i for $i = 1, 2$ and $j = 1, 2, \dots, n$, and $x_{ij} = 1$ if the prediction is correct, and $x_{ij} = 0$ otherwise. When the number of folds is k , the dependency score between the accuracies resulting from two replications is the expected proportion of $x_{1j} = x_{2j}$ denoted by ρ_k .

The partition of a data set into disjoint folds is random. The proportion of $x_{1j} = x_{2j}$ for a pair of replications is not constant, and every possible partition have the same occurring probability. The dependency score ρ_k depends not only on data set and classification algorithm, but also on the

number of folds. The dependency score always equals one when $k = n$, while it is generally less than one when $k < n$.

The dependency score represents the probability of having both correct or both wrong predictions on the same instance in two replications. When the predictions of an instance in two replications are both correct or both wrong, it does not necessarily mean that the two predictions are dependent. Consider classification algorithm 1NN. When an instance is classified by different instances in two replications, these two predictions are independent and can be both correct. Since accuracy is an aggregation of zero-one loss, the dependency score is appropriate for measuring the correlation between the accuracies obtained from two replications.

Example 3. Suppose that a data set D has four instances e_1 through e_4 . When the number of folds is two, the three possible partitions are $\{\{e_1, e_2\}, \{e_3, e_4\}\}$, $\{\{e_1, e_3\}, \{e_2, e_4\}\}$, and $\{\{e_1, e_4\}, \{e_2, e_3\}\}$. Suppose that the testing results by a classification algorithm L for the three partitions are $\{1, 0, 0, 1\}$, $\{1, 0, 1, 1\}$, and $\{1, 0, 1, 0\}$. The proportions of $x_{1j} = x_{2j}$ between partitions (1, 2), (1, 3), and (2, 3) are $3/4$, $2/4$, and $3/4$, respectively. When 2-fold cross validation is applied on this data set twice for evaluating algorithm L , the possible combinations for the three partitions are nine. The dependency score for 2-fold cross validation is

$$\rho_2 = (1 + 1 + 1 + 3/4 \times 2 + 2/4 \times 2 + 3/4 \times 2)/9 = 7/9.$$

For every two replications of k -fold cross validation, the dependency score ρ_k can be estimated as $r_k = (b_{11} + b_{00})/n$. We will have $r_k = 1$ when $x_{1j} = x_{2j}$ for all j , and $r_k = 0$ when $x_{1j} \neq x_{2j}$ for all j . If the two predictions for instance e_j are independent, then

$$\begin{aligned} P(x_{1j} = x_{2j}) &= P(x_{1j} = x_{2j} = 1) + P(x_{1j} = x_{2j} = 0) \\ &= P(x_{1j} = 1)P(x_{2j} = 1) + P(x_{1j} = 0)P(x_{2j} = 0) \\ &= p^2 + (1 - p)^2. \end{aligned}$$

When the predictions in two replications are independent, the dependency score equals $p^2 + (1 - p)^2$. The value of p depends on both data set and classification algorithm. This dependency score is normalized as $\rho'_k = (\rho_k - q)/(1 - q)$ where $q = p^2 + (1 - p)^2$. The value of ρ'_k will be between 0 and 1 for all possible data sets and classification algorithms when ρ_k is between q and 1. It is extremely unlikely that the predictions in two replications are negatively correlated. The normalized dependency score can thus be used to evaluate the strength of the dependence level between two replications of k -fold cross validation.

Theorem 4. When the large-sample conditions $n\rho_k \geq 5$ and $n(1 - \rho_k) \geq 5$ hold for a data set with n instances, the estimated normalized dependency score $r'_k = (r_k - q)/(1 - q)$ between two replications of k -fold cross validation can be assumed to follow a normal distribution with mean $(\rho_k - q)/(1 - q)$ and variance $\frac{\rho_k(1 - \rho_k)}{n(1 - q)^2}$.

Proof. Define binary random variable $y_j = I(x_{1j} = x_{2j})$ for instance e_j for $j = 1, 2, \dots, n$. Then $P(y_j = 1) = P(x_{1j} = x_{2j}) = \rho_k$. Since instances are collected by the simple random sampling, the y_j for $j = 1, 2, \dots, n$ are independent. Random variable $\sum_{j=1}^n y_j = b_{11} + b_{00}$ will follow a binomial distribution with parameters n and ρ_k . When the large

sample conditions $n\rho_k \geq 5$ and $n(1 - \rho_k) \geq 5$ hold, by the normal approximation of binomial distribution, $\sum_{j=1}^n y_j$ can be assumed to follow a normal distribution with mean $n\rho_k$ and variance $n\rho_k(1 - \rho_k)$. The estimate of dependency score $r_k = (b_{11} + b_{00})/n$ can thus be assumed to follow a normal distribution with mean ρ_k and variance $\rho_k(1 - \rho_k)/n$. Multiplying a random variable following a normal distribution by a nonzero constant is also normally distributed. The estimated normalized dependency score $r'_k = (r_k - q)/(1 - q)$ thus follows a normal distribution with mean

$$E(r'_k) = E(r - q)/(1 - q) = (\rho_k - q)/(1 - q).$$

and variance

$$Var(r'_k) = \frac{Var(r_k - q)}{(1 - q)^2} = \frac{Var(r_k)}{(1 - q)^2} = \frac{\rho_k(1 - \rho_k)}{n(1 - q)^2}.$$

□

The hypothesis for testing the dependency score between two replications is $H_0: \rho'_k \leq r_0$ for any given dependency level $\rho_0 > 0$. Since the accuracies obtained from two replications are both estimates of the actual accuracy p , they are pooled together to obtain sample accuracy $\bar{p} = \sum_{i=1}^2 \sum_{j=1}^n x_{ij}/2n$. The statistic for testing this hypothesis is

$$z = \frac{r'_k - \rho_0}{\sqrt{\frac{\bar{p}(1 - \bar{p})}{n(1 - \bar{q})^2}}},$$

where $\bar{q} = \bar{p}^2 + (1 - \bar{p})^2$ and $\bar{p} = \rho_0(1 - \bar{q}) + \bar{q}$. When significance level is α , the normalized dependency score ρ'_k is considered to be larger than ρ_0 if $z \geq z_\alpha$.

Example 4. Suppose that a data set D has four instances e_1 through e_4 . The testing results by a classification algorithm L for the two replications of 2-fold cross validation are $\{1, 0, 0, 1\}$ and $\{1, 0, 1, 1\}$. Let $\rho_0 = 0.4$, and hence $r_2 = 3/4$, $\bar{p} = 5/8$, $\bar{q} = 17/32$, $\bar{p} = 0.4 \times (1 - 17/32) + 17/32 = 0.72$, and $r'_2 = (3/4 - 17/32)/(1 - 17/32) = 7/15$. The statistic for testing hypothesis $H_0: \rho'_2 \leq 0.4$ is

$$z = \frac{7/15 - 0.4}{\sqrt{\frac{0.72 \times (1 - 0.72)}{4 \times (1 - 17/32)}}} = 0.2789,$$

and its corresponding p -value is 0.3902. When significance level α is 0.05, the normalized dependency score of 2-fold cross validation is not significantly larger than 0.4.

It is impossible to know whether the large-sample conditions $n\rho_k \geq 5$ and $n(1 - \rho_k) \geq 5$ are satisfied or not for the testing results of a data set because ρ_k is unknown. The value of $n\rho_k$ specifies the expected number of instances that have both correct or both wrong predictions in two replications. The large-sample conditions for a data set are thus replaced by $b_{11} + b_{00} \geq 5$ and $b_{10} + b_{01} \geq 5$; i.e., the numbers of instances with the same and different prediction outcomes are both larger than or equal to five.

Two classification algorithms are chosen for analyzing the dependency relationships among the replications of k -fold cross validation. Naïve Bayesian classifier calculates the classification probability of every possible class value for a testing instance, and assigns the one that has the largest probability

TABLE 3
The Basic Characteristics of Data Sets

Data set	No. of instances	No. of attributes	No. of class values
Australian	690	14	2
Balance-scale	625	4	3
Cleve	303	13	5
Climate	540	20	2
Ecoli	336	8	8
Glass	214	10	7
Haberman	306	3	2
Heart	270	13	2
Liver	345	7	2
Magic	19,020	11	2
Page-blocks	5,473	10	5
Pendigits	10,992	16	10
Pima	768	8	2
Segmentation	2,310	19	7
Spambase	4,601	57	2
Statlog	6,435	36	6
Transfusion	748	4	2
Vehicle	946	18	4
White-wine	4,898	11	11
Yeast	1,484	8	10

to be the predicted class value. This algorithm is more suitable for processing discrete attributes. The second one is support vector machine that derives a hyperplane for every two class values, and this hyperplane will be used for classification. Deriving the hyperplanes by optimization is more straightforward when attributes are continuous.

5 EXPERIMENTAL RESULTS

Twenty data sets are downloaded from the UCI data repository [17] to perform the tests of independence and the normalized dependency score introduced in the previous section, and the basic characteristics of data sets are summarized in Table 3. The number of instances is between 214 and 19,020, the number of attributes is between 3 and 57, and the number of class values is between 2 and 11. The characteristics of an ordinal data set should be in these ranges. The data sets that cannot satisfy the large-sample conditions are also excluded. The number of folds is set to be 2, 5, and 10 for identifying whether the number of folds will have impact on the dependency level among the accuracies resulting from various replications. Naïve Bayesian classifier and support vector machine are provided by software R with default settings, and these two algorithms are abbreviated as NBC and SVM, respectively.

The critical chi-squared value corresponding to significance level $\alpha = 0.01$ with one degree of freedom is $\chi^2_{0.01,1} = 6.6349$. When the test statistic $\chi^2 = n\bar{\phi}$ is larger, its corresponding p -value will be closer to zero. The statistics χ^2 of the independence test for naïve Bayesian classifier and support vector machine are shown in Table 4 in which every test statistic is far larger than the critical value 6.6349. This indicates that it is absolutely inappropriate to assume that the accuracies obtained from various replications are independent.

Since the test statistics shown in Table 4 are much larger than the critical value 6.6349, their corresponding p -values are all very close to zero. The test statistic for any given data set is generally getting larger as the number of folds increases.

TABLE 4
The Statistics χ^2 of the Independence Test for (a) Naïve Bayesian Classifier and (b) Support Vector Machine

(a)	Data set	2 folds	5 folds	10 folds
	Australian	478.36	532.94	599.52
	Balance-scale	264.61	467.50	564.28
	Cleve	238.78	261.65	295.88
	Climate	265.52	361.71	431.40
	Ecoli	151.21	253.74	257.70
	Glass	120.03	120.27	153.79
	Haberman	221.53	223.37	241.14
	Heart	140.57	192.10	215.63
	Liver	61.84	176.61	216.60
	Magic	18,140.61	18,614.08	18,672.01
	Page-blocks	4,113.57	4,451.63	4,641.96
	Pendigits	9,663.73	10,321.58	10,461.30
	Pima	477.87	551.51	648.84
	Segmentation	1,521.44	1,942.28	2,112.20
	Spambase	4,032.75	4,315.19	4,374.04
	Statlog	6,145.43	6,293.40	6,344.70
	Transfusion	504.63	528.90	611.63
	Vehicle	511.30	630.04	712.60
	White-wine	3,744.79	4,273.57	4,471.19
	Yeast	798.84	1,173.07	1,289.20
(b)	Data set	2 folds	5 folds	10 folds
	Australian	590.29	682.01	682.01
	Balance-scale	339.53	514.68	532.35
	Cleve	123.03	237.00	254.73
	Climate	501.13	497.71	497.71
	Ecoli	139.61	245.53	293.15
	Glass	124.54	168.93	181.70
	Haberman	219.95	251.06	290.62
	Heart	180.47	178.53	233.44
	Liver	71.01	203.79	256.56
	Magic	16,442.92	17,424.69	17,835.15
	Page-blocks	3,997.44	4,727.87	4,910.47
	Pendigits	8,183.43	8,827.27	9,107.62
	Pima	453.70	556.52	573.61
	Segmentation	1,652.68	2,077.47	2,112.44
	Spambase	3,390.72	4,023.57	4,164.45
	Statlog	5,855.31	6,072.71	6,209.61
	Transfusion	484.69	535.75	651.27
	Vehicle	289.94	394.63	573.40
	White-wine	3,165.60	3,762.05	4,179.40
	Yeast	925.99	1,214.14	1,299.22

It is thus more inappropriate to assume that the accuracies resulting from various replications are independent for large number of folds. However, the test statistics in Table 4 cannot be used to measure the strength of the dependency level between the accuracy estimates for a pair of replications. The test of the normalized dependency score is designed for this purpose.

In order to know whether the accuracy estimates obtained from various replications are highly dependent, the threshold ρ_0 is set to be 0.80, 0.85, 0.90, or 0.95. The results of the tests for the normalized dependency score are summarized in Table 5 for significance level $\alpha = 0.05$. In both algorithms, more than half of the 20 data sets have a normalized dependency score larger than 0.80 when the number of folds is ten, and the number of significances increases as the number of folds increases. The number of folds thus have a large impact on the dependency level among the accuracy estimates in various replications. The results indicate that the positive

TABLE 5
The Number of Data Sets That Have Normalized Dependency Score Significantly Larger Than ρ_0 When $\alpha = 0.05$

Algorithm	Number of folds	ρ_0			
		0.80	0.85	0.90	0.95
NBC	2	6	5	4	2
	5	15	9	5	4
	10	18	14	11	5
SVM	2	8	4	2	0
	5	16	10	6	3
	10	19	15	11	5

correlation among the accuracy estimates obtained from various replications is getting strong as the number of folds increases.

6 DISCUSSION

The analytical results given in Section 3 show that the probability for the predictions of the same instances by 1NN to be dependent in two replications of k -fold cross validation is getting large as the number of folds increases. The statistical tests performed in Section 5 also demonstrate that the dependency level of the accuracies obtained from various replications exhibits the same pattern for naïve Bayesian classifier and support vector machine. These may suggest that it is appropriate to perform k -fold cross validation several times when k is small. The number of folds is generally set to be 2, 5, or 10 in exploring the characteristics of k -fold cross validation [2], [13], [14], [18]. This section will study 5×2 , 2×5 , and 1×10 cross validation to draw conclusions for using k -fold cross validation in performance evaluation.

Let a data set D contain n instances collected by the simple random sampling. When the number of folds $k = n$, an instance is classified by the model induced from the same $n - 1$ instances in various replications. The accuracies obtained from the replications of leave-one-out cross validation are therefore fully dependent. Let M_1 and M_2 be the respective models induced in replications 1 and 2 of k -fold cross validation for classifying instance e . If the training sets for M_1 and M_2 have a lot of instances in common, it is likely that both models will have the same prediction on instance e . This implies that when the training sets for M_1 and M_2 have more instances in common, the probability for them to have the same prediction on an instance is larger. Analyzing the common training instances for the models induced in various replications can help us to identify the dependency level between the accuracies obtained from two replications.

Theorem 5. *In a data set having n instances, let T be the number of common training instances for inducing models to classify an instance in two replications of k -fold cross validation. Then $E(T) = \frac{n^2(k-1)^2}{(n-1)k^2}$.*

Proof. Assume without loss of generality that every fold has m instances, and hence $n = k \times m$. Let F_{ij} represent the j^{th} fold in replication i . Suppose that an instance e is assigned to F_{1u} and F_{2v} for some u and v . Then the training set for inducing the model to classify instance e in the first and the second replications are $\cup_{j \neq u} F_{1j}$ and $\cup_{j \neq v} F_{2j}$, respectively. Each one of the two training sets has $n - m$ instances. Since

the partition of instances into folds is random in every replication, and the partitions of every pair of replications are independent, the probability for an instance in $\cup_{j \neq u} F_{1j}$ to be assigned to $\cup_{j \neq v} F_{2j}$ is $(n - m)/(n - 1)$. The expected number of instances in $\cup_{j \neq u} F_{1j}$ that will be in $\cup_{j \neq v} F_{2j}$ is

$$E(T) = (n - m) \times \frac{n - m}{n - 1} = \frac{(n - m)^2}{n - 1} = \frac{n^2(k - 1)^2}{(n - 1)k^2}. \quad \square$$

By Theorem 5, the expected number of common training instances for any given data set increases as the number of folds is getting larger. The expected proportion of common training instances between two replications is

$$\frac{E(T)}{n - 1} = \frac{n^2(k - 1)^2}{(n - 1)^2 k^2}.$$

When $k = n$, $E(T)/(n - 1) = 100\%$ indicates that the training set for each instance is exactly the same in every replication of leave-one-out cross validation. When n is large, $E(T)/(n - 1) \approx (k - 1)^2/k^2$. The expected common proportions for $k = 2, 5$, and 10 are approximately 25, 64, and 81 percent, respectively for large data sets. The common training proportion for $k = 10$ is more than three times of that for $k = 2$. The common training proportion could explain why the dependency level is getting stronger when the number of folds increases.

In the study proposed by Wong and Yang [15] for analyzing the dependency among the fold accuracies obtained in a single run of k -fold cross validation, the dependency is due to the cross validation between two distinct instances. The number of folds thus does not have impact on the independence assumption of fold accuracies. This study investigates the predictions of the same instance and the predictions of distinct instances in various replications of k -fold cross validation. The number of folds becomes critical in determining the dependency level among the accuracy estimates obtained from various replications.

If ten fold accuracies is going to be collected for statistical inference, which alternative among 5×2 , 2×5 , and 1×10 cross validation should be adopted? An instance plays the role of training for five times and the role of testing for five times in 5×2 cross validation. These two values for an instance are eight and two in 2×5 cross validation, and nine and one in 1×10 cross validation. When the training and testing times for processing an instance are identical, these three evaluation alternatives will have the same computational cost.

Suppose that a classification algorithm with actual accuracy p is tested on a data set that has $n = k \times m$ instances. Let $\bar{p}_{ij}^\lambda$ be the resulting accuracy of fold j in the i^{th} replication of alternative λ for $\lambda = 5 \times 2, 2 \times 5$, and 1×10 . By Section 2, when an evaluation alternative is repeated, $\bar{s}^2$ is a more appropriate estimate of sample variance, and hence it will be used for variance estimation. Let $\bar{s}_\lambda^2 = \sum_{i=1}^l [\sum_{j=1}^k (\bar{p}_{ij}^\lambda - \bar{x}_{\lambda i})^2 / (k - 1)] / l$ be the sample variance obtained from the l replications of alternative λ , where $\bar{x}_{\lambda i} = \sum_{j=1}^k \bar{p}_{ij}^\lambda / k$. Though the fold accuracies obtained from various replications are dependent, $\bar{s}_\lambda^2/k$ is an unbiased point estimator of the variance of mean accuracy regardless of the evaluation alternative. Therefore, when $\bar{s}_\lambda^2/k$ is used to estimate the variance of mean accuracy, and the testing results of every fold satisfy

the large-sample conditions, the numbers of folds and replications are both arbitrary. Since the dependency level among the fold accuracies obtained from various replications is getting strong as the number of replications increases, it is inappropriate to say the mean accuracy resulting from a larger number of replications will be more reliable. That is the main reason why Dietterich [9] suggested that 2-fold cross validation should not be repeated for more than five times.

Since the variance estimates $\bar{s}_\lambda^2/k$ for the three evaluation alternatives are all unbiased point estimator of $p(1-p)/n$, their adoption depends on which alternative is more powerful in identifying the superior one between two classification algorithms. When the actual accuracy p is estimated by the average of fold accuracies, this estimate is a sample mean instead of a sample proportion. If variance $p(1-p)/n$ is estimated by $\bar{s}_\lambda^2/k$, the test statistic will be the t value. The degrees of freedom of $\bar{s}_\lambda^2$ will be used to determine the degrees of freedom of t value. The degrees of freedom of $\bar{s}_\lambda^2$ for $\lambda = 5 \times 2, 2 \times 5$, and 1×10 are 5, 8, and 9, respectively. A larger degrees of freedom is more powerful in identifying the superior algorithm.

Based on the above discussion, the procedure for applying k -fold cross validation is suggested as follows. If the testing results of every fold in 10-fold cross validation can satisfy the large-sample conditions, then it should be performed only once. By Wong and Yang [15], those fold accuracies can be assumed to be independent for statistical inference. The experimental results given in Table 5 indicate that the dependency level among the accuracy estimates obtained from various replications is generally strong in 10-fold cross validation, and hence it should not be repeated. If the testing results of some folds in 10-fold cross validation violate the large-sample conditions, reduce the value of k until the large-sample conditions hold for every fold. Then this k -fold cross validation can be repeated l times such that $l \times k$ is approximately equal to 10. When k -fold cross validation is repeated, the sample variance should be calculated by $\bar{s}^2$.

An imbalanced data set generally has two class values, and the goal is to achieve high prediction accuracy on the one with few instances. Since the evaluation measures for imbalanced data sets are also based on zero-one loss, Definition 1 given in Section 2 can also be used to delineate the dependency relationships between the predictions in two replications of k -fold cross validation. The conclusions given in this section are therefore applicable for imbalanced data sets, while the large-sample conditions should be adjusted according to the measure used for performance evaluation.

7 CONCLUSIONS

Whether k -fold cross validation should be repeated is essential in obtaining reliable accuracy estimates for evaluating the performance of classification algorithms. This study first presents three ways to calculate variance estimates, and addressed that the fold accuracies resulting from various replications should not be assumed to be independent in variance estimation. Then the k -nearest neighbors with $k = 1$ is employed to explore the dependency relationships among the predictions of instances in various replications. The analytical results of the k -nearest neighbors with $k = 1$ show that

the dependency level is getting strong as the number of folds increases.

Since analyzing the dependency level for the other classification algorithms is not as easy as for the k -nearest neighbors with $k = 1$, two statistical methods are proposed to test whether the accuracy estimates in various replications are independent or not. The experimental results on 20 data sets for naïve Bayesian classifier and support vector machine demonstrate that the independence assumption for the accuracy estimates obtained from various replications is absolutely inappropriate. Similar to the analytical results obtained in the k -nearest neighbors with $k = 1$, the dependency level will be getting strong as the number of folds becomes large.

Suppose that there are three possible alternatives for performance evaluation: repeat 2-fold cross validation five times, repeat 5-fold cross validation twice, and perform 10-fold cross validation once. The last alternative should be the first choice. If the large-sample conditions do not hold for some fold in 10-fold cross validation, then repeating 5-fold cross validation twice is the next option. Repeating 2-fold cross validation five times will be adopted only when the first two alternatives both violate the large-sample conditions in some folds.

Though k -fold cross validation can also be applied on the data sets for numeric prediction, the evaluation of classification algorithms is completely different from that of the algorithms for numeric prediction. The definition of dependence for zero-one loss proposed in this study is not applicable for squared-error loss. It should thus be desirable to know whether the repetition of k -fold cross validation can reduce the variance estimate for measuring prediction error such as mean absolute error and mean squared error. It may be worthwhile to explore whether the number of folds have the same impact as in evaluating classification algorithms.

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